MaSyMoS:
Finding hidden treasures in model repositories

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SYSTEMS BIOLOGY
BIOINFORMATICS
ROSTOCK

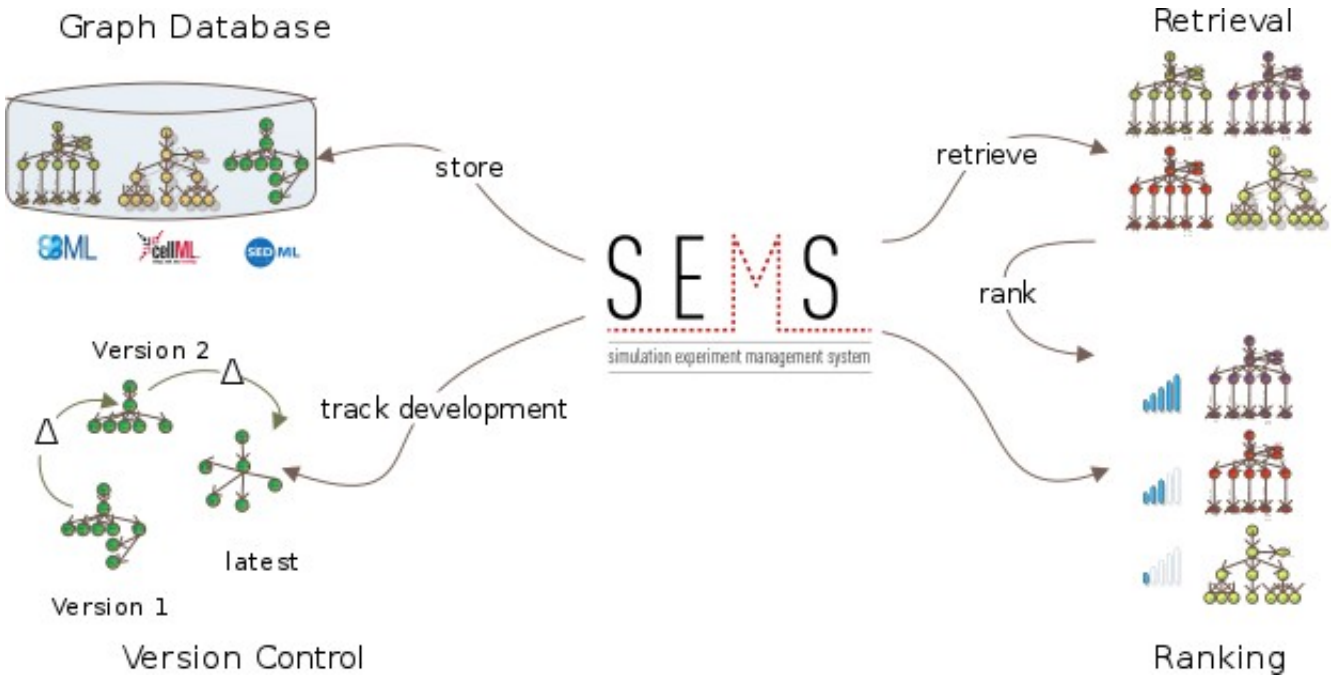
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"We've been hearing a common theme from the academic community– researchers are having difficulty managing and accessing their data. It seems to be an ongoing problem for research scientists"

Nature Blogs (2014) *Nine Worrying Stats on the Effect of Poor Scientific Data Management.*



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Model

Entities, network of reactions, math

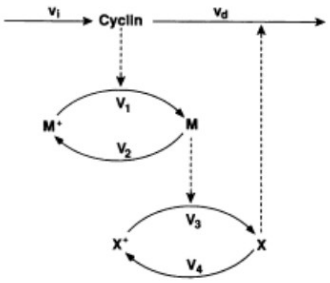


Fig: Goldbeter (1991), http://www.ncbi.nlm.nih.gov/pubmed/1833774

Annotations

Compartment: Cell GO:0005623
Publication: Goldbeter PMID:1833774
M = inactive CDCD2 Kinase: UniProt:CDK1a_XENIA

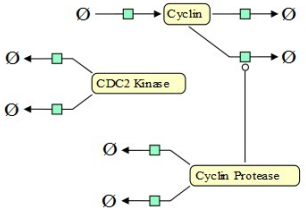


Fig.: BioModels Database

Protocols

Behavior: Oscillation
TEDDY_0000006
Algorithm: Gillespie
KiSAO:000029

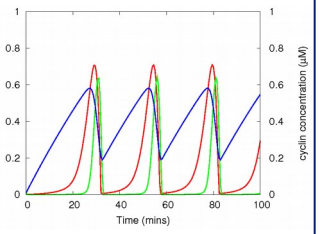
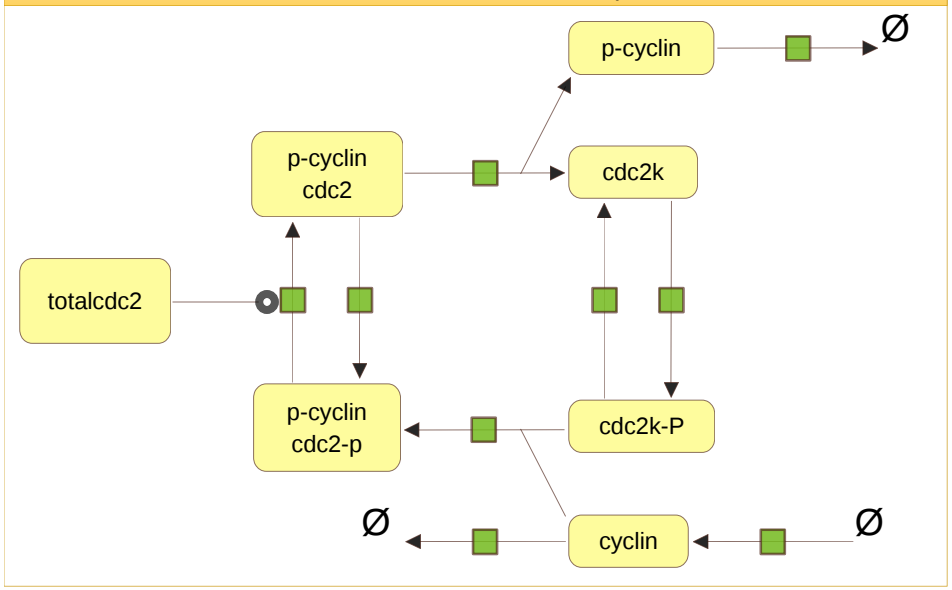


Fig.: BioModels Database

combine the computational modeling in biology network

Modeling the cell division cycle: cdc2 and cyclin interactions, John J. Tyson, 1991. SBGN Process Description



Which are the most frequently used GO annotations in my model set?

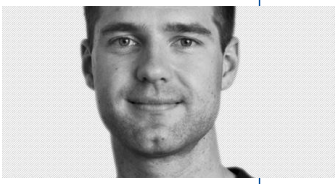
Which models are annotated with ‘Adenosine tri-phosphate’?

How many models are available for this PubMed entry?

Give me all the files I need to run this simulation study.

Find good candidates for features describing my model set. (Alm et al 2014)

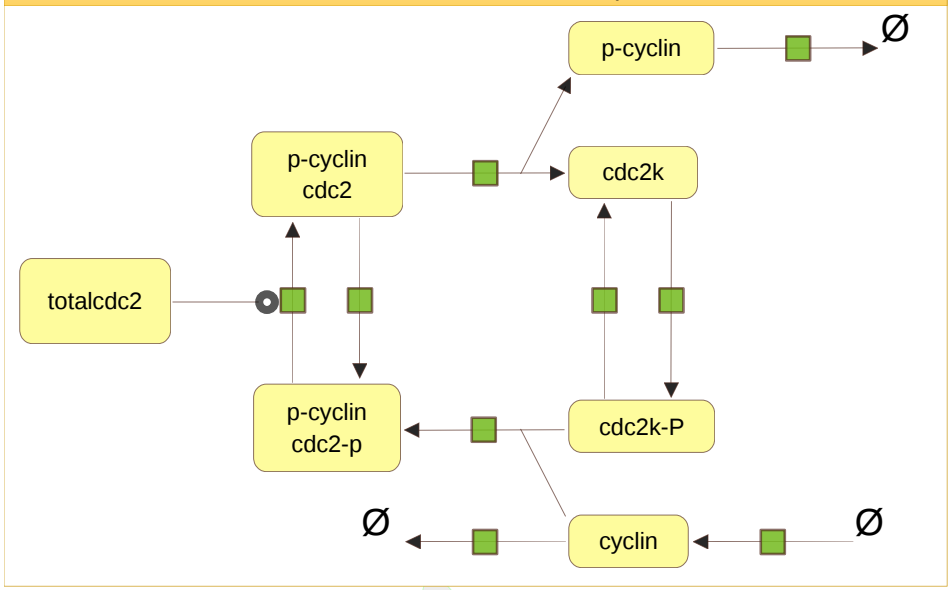
Which models contain reactions with 'ATP' as reactant and 'ADP' as product?



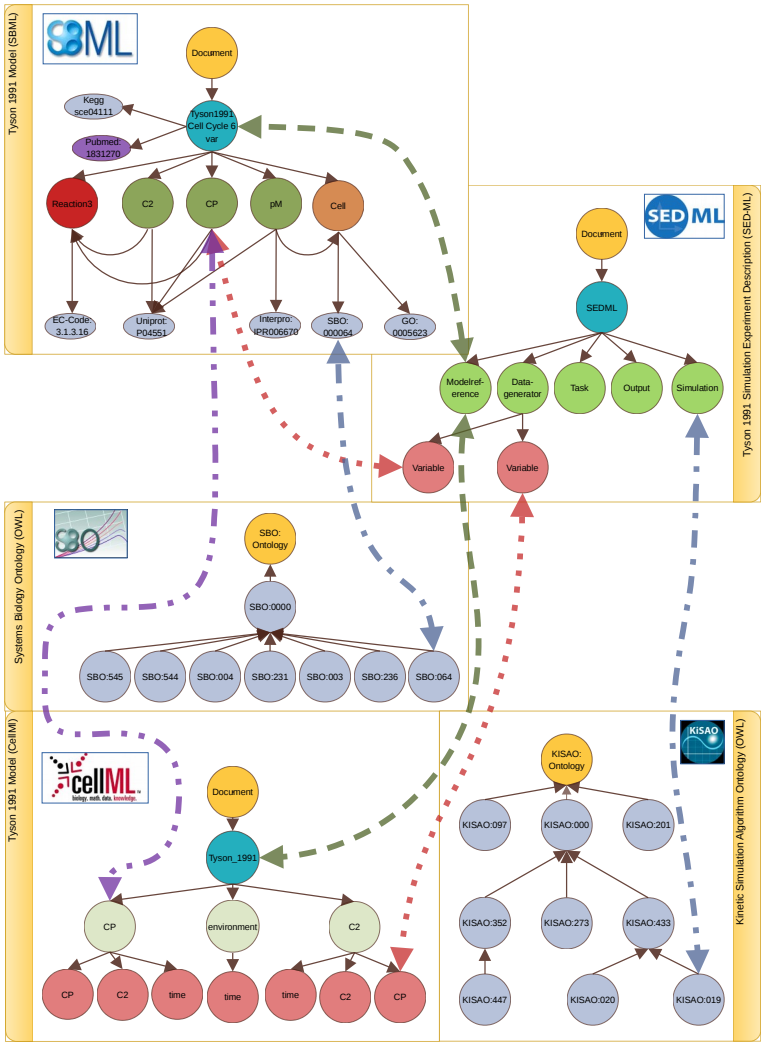
Ron Henkel

Figure courtesy Ron Henkel, https://peerj.com/preprints/376/

Modeling the cell division cycle: cdc2 and cyclin interactions, John J. Tyson, 1991. SBGN Process Description







Figures courtesy Ron Henkel, https://peerj.com/preprints/376/

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SEMS
simulation experiment management system

Try it out: https://sems.uni-rostock.de/projects/masymos/



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